Cloud Operations and Management

Executive Summary and Assessment

Cloud Migration Strategy for Al Dar Furnishings

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1.0 Document Control

1.1 Identity

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1.3 Author

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1.4 Document Purpose

This report evaluates the current on-premise infrastructure at Al Dar Furnishings, a Dubai-based retailer, and proposes a cloud-first migration strategy on Microsoft Azure. It covers infrastructure design with Infrastructure as Code, AI-driven security, risk and compliance, and recommendations for future direction.

2.0 Executive Summary

Al Dar Furnishings is a Dubai-based home furnishing retailer with 10 showrooms spread across the UAE, Qatar, and Bahrain, 4 warehouses, and around 165 staff. Al Dar was recently hit by a ransomware attack. It brought down their on-premise infrastructure completely. The primary reason for the attack was identified as lack of multi-factor authentication, no endpoint detection or patching, branch firewalls running without licenses, no disaster recovery plan, and no data sovereignty controls across three countries.

This report puts forward a cloud-first migration to Microsoft Azure, replacing the legacy environment with SaaS, PaaS, and IaaS services as defined by Mell and Grance (2011). The proposed solution includes:

- Microsoft 365 and Dynamics 365 for productivity and ERP
- Entra ID with Intune for identity and device management
- Cato Networks SASE for secure site connectivity
- Microsoft Defender for AI-driven threat detection
- Terraform for infrastructure provisioning

The migration would reduce the on-premise footprint from 20 VMs down to just 2 workloads, move roughly 70% of infrastructure onto SaaS and PaaS with built-in redundancy, and close all seven identified gaps. Having said that, the heavy use of Microsoft cloud is a real concern, and this report treats vendor concentration as a risk to be actively managed through portable IaC and annual reviews.

3.0 Current State Assessment

An assessment was conducted for the existing infrastructure covering the network, compute, identity, data, applications, and security layers.

3.1 Infrastructure Discovery

The following tables summarise the key components discovered.

3.1.1 Network

Component	Detail
HQ Perimeter Firewall	FortiGate FG-300D
HQ Internal Firewall	Cisco ASA-5512
Core Switches	Cisco Catalyst 3650-24P x 2
Branch Firewalls	FortiGate 60F x 10 (unlicensed)
WAN Topology	Hub-spoke VPN across all sites (10Mb leased line)
Wireless	Cisco Aironet and Aruba (mixed)

3.1.2 Compute

Component	Detail
Hypervisor	VMware (versions 5 through 7)
Virtual Machines	~20 VMs
Operating Systems	Windows Server 2016, Windows Server 2008 (EOL)

3.1.3 Identity and Access

Component	Detail
Directory Services	On-premise Active Directory
Endpoint Management	Group Policy
Domain Membership	All Windows machines domain-joined

3.1.4 Data

Component	Detail
Database	SQL Server 2016 Standard
Document Management	On-premise DMS

Backup	Veeam to local SAN

3.1.5 Applications

Application	Purpose	Hosting
Microsoft Dynamics FnO	ERP (finance, supply chain, operations)	On-premise
IN Concert	Customer service call centre	On-premise
Websites	Consumer-facing B2C and D2C channels	External
Field Services	On-site service delivery	Independent

3.1.6 Security

Component	Detail
Endpoint Protection	Kaspersky (signature-based)
Branch Firewall Security Services	Inactive (expired licenses)
Recent Incident	Ransomware attack by external threat group

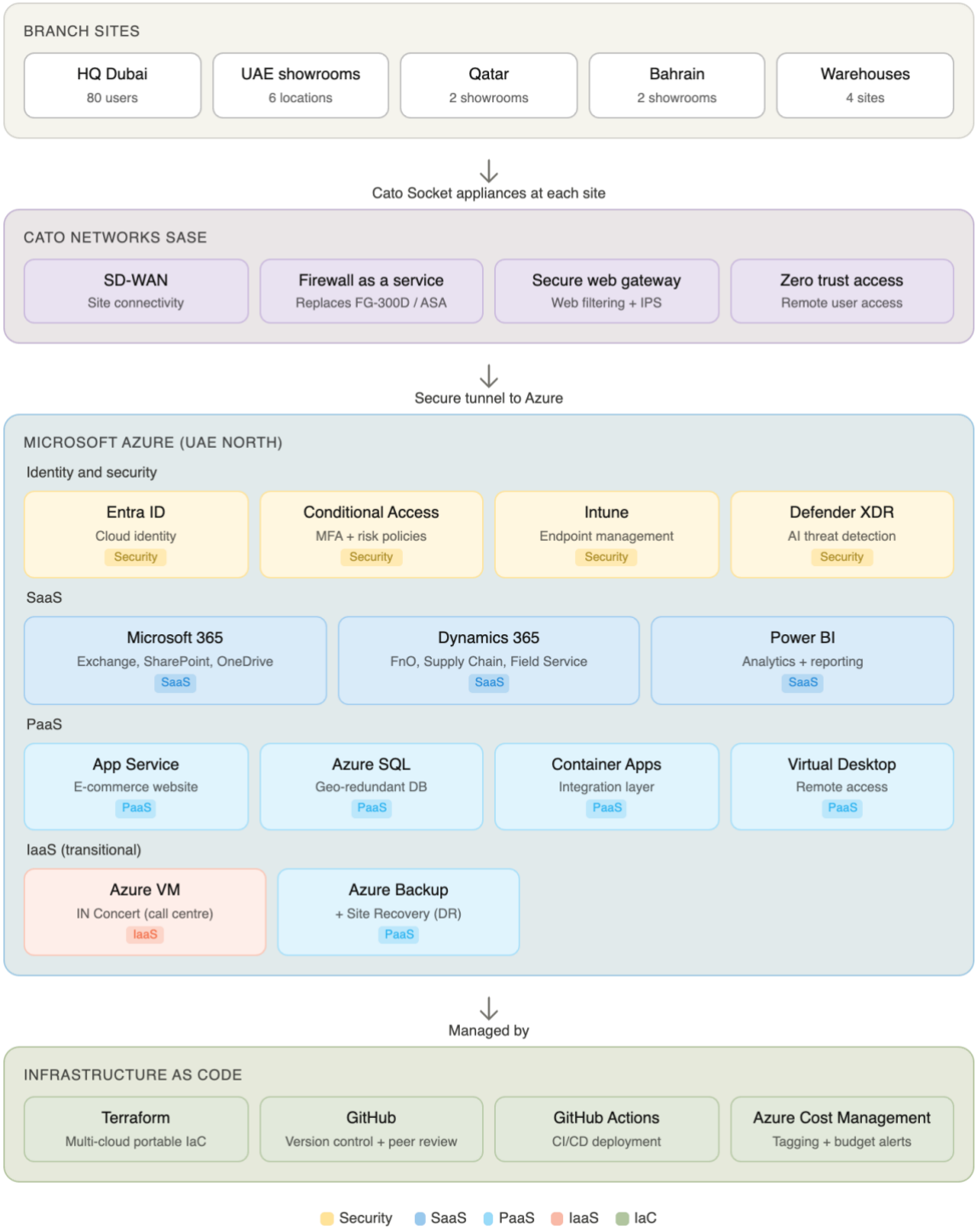
3.2 Gap Analysis

Based on the infrastructure discovery, the following functional, security, and compliance gaps were identified and categorised against the NIST Cybersecurity Framework domains of Identify, Protect, Detect, Respond, and Recover (NIST, 2018). Each gap is referenced in Section 4.0 where the corresponding mitigation is proposed.

Type	Gap	Impact
Security	No business continuity or disaster recovery	Local-only backups; no recovery from site-level incidents
Security	No centralised identity governance	Password-only auth across 10 sites; credential attacks unmitigated
Security	No endpoint detection, patching, or vulnerability management	Unpatched systems and signature-only AV enabled ransomware compromise and lateral movement
Compliance	No data sovereignty or regulatory controls	Three-country operation with no data residency enforcement
Functional	No modern endpoint management	Group Policy limited to on-network; no MDM or compliance reporting

Functional	No scalable infrastructure for growth	EOL hypervisors and OS bound to single site
Functional	No standardised deployment or change control	Manual changes with no audit trail

4.0 Proposed Strategy



The strategy is to move Al Dar Furnishings off its on-premise infrastructure and onto Microsoft Azure, directly addressing the gaps from Section 3.2. Upgrading what already exists was considered, but that only fixes the immediate security holes and does nothing about scalability or the risk of everything sitting in one building. Gupta (2024) makes the case that organisations on ageing infrastructure get more value from cloud transformation than from patching.

Azure won out over AWS and Google Cloud because the business is already deep in the Microsoft stack. The vendor dependency is a concern (Dunbar, 2024), but the integration savings are hard to ignore. Terraform is used instead of Bicep for portability (Morris, 2020). The migration runs in three phases: identity and security first, then applications, then infrastructure and decommission.

4.1 Workload Migration Plan

The table below maps each current on-premise workload to its proposed cloud target.

Current Workload	Target	Justification
Domain controllers + DNS + DHCP	Microsoft Entra ID (decommission)	Removes identity SPOF; enables MFA
Exchange Server (mailbox + transport)	Microsoft 365 Exchange Online	Built-in redundancy; reduces overhead
File and print server	SharePoint Online, OneDrive, Universal Print	Remote access across all sites
Dynamics FnO (app servers + AOS)	Dynamics 365 Finance and Operations	SaaS with built-in HA and DR
SQL Server (Dynamics DB)	Managed by Dynamics 365 (decommission)	Removes patching and HA burden
DMS server	SharePoint Online	Consolidates into M365 ecosystem
E-commerce web servers x2	Azure App Service	Auto-scaling; PaaS reduces management
SQL Server (e-commerce DB)	Azure SQL Database (geo-redundant)	Built-in HA; addresses single-instance risk
IN Concert (app + DB)	Azure VM (short-term), CCaaS (long-term)	Legacy app; IaaS as transitional step

Reporting server	Power BI	Self-service analytics; eliminates infra
Integration / middleware	Azure Container Apps	Managed orchestration; event-driven scaling
Remote desktop gateway	Azure Virtual Desktop	Secure access without VPN dependency
Veeam backup server	Azure Backup + Site Recovery	Offsite replication; addresses DR gap
Kaspersky mgmt + vCenter + WSUS	Decommission	Functions absorbed by cloud services

Roughly half the workloads end up on SaaS, which takes a lot of day-to-day management off the IT team's plate. Another five move to PaaS where high availability comes built in. IN Concert is the awkward one: it is a legacy call centre application with no obvious cloud-native replacement, so it sits on an Azure VM for now while the business evaluates alternatives (Gupta, 2024). The infrastructure VMs like domain controllers and vCenter simply go away once their functions are handled by cloud services.

4.2 Network and Connectivity

The network architecture is replaced with a cloud-native SASE model as outlined below.

Current State	Proposed State
Hub-spoke VPN over 10Mb leased line	Cato Networks SASE with SD-WAN
FG-300D / ASA-5512 at HQ	Replaced by Cato PoP connectivity
FortiGate 60F (unlicensed) at branches	Replaced by Cato Socket appliances

Cato is a cloud-native SASE platform that bundles SD-WAN and security together. Each site gets a Cato Socket appliance connecting to the nearest Point of Presence, which means the unlicensed FortiGate 60F units at branches are no more required. We also evaluated Zscaler and Palo Alto Prisma, but Cato was chosen because it covers both networking and security in one platform. This would matter when the IT

team is small (Cato Networks, 2024). The obvious downside is that this creates a single-vendor dependency for the entire WAN.

4.3 Identity and Access Management

Identity and endpoint management are migrated from on-premise to cloud-native services.

Current State	Proposed State
On-premise Active Directory	Microsoft Entra ID
Group Policy	Microsoft Intune

During the transition, Azure AD Connect keeps on-premise and cloud identities in sync. This hybrid period is necessary but not without risk: policies can end up inconsistent between the two directories, so the cutover to cloud-only should not happen until every application dependency on on-premise AD has been removed. Intune takes over from Group Policy for device compliance, patching, and app deployment across all 165 machines. Conditional Access will enforce MFA and allow access from devices with a healthy posture. This will cover the credential-based attack that made the ransomware possible in the first place (Microsoft, 2023).

4.4 Infrastructure as Code

Terraform handles the deployment side of things. The setup will be split into different modules, covering networking, compute, data, and identity. Each module will have separate state files for production and staging. This ensures changes in one do not break the other. All of it sits in Git. Any change goes through peer review before it gets deployed through GitHub Actions. On the cost side, every resource is tagged from the start and Azure Cost Management is configured to flag spend that looks off, which Feitosa et al. (2024) recommend as a baseline practice for any IaC-managed environment.

Financially, this moves the business from buying and maintaining hardware to a predictable monthly spend. PaaS services like App Service and Azure SQL also bring auto-scaling, which means the e-commerce site can handle traffic spikes without the IT team manually spinning up capacity.

5.0 Security, Risk, and Compliance

5.1 Threat Landscape and Ransomware Lessons

The ransomware attack revealed multiple control failures in the environment. The absence of MFA allowed compromised credentials to be used without challenge. The absence of EDR meant lateral movement across the network went undetected. Unpatched systems provided known vulnerabilities for the attackers to exploit, and a flat network with no segmentation allowed the attack to spread unchecked. Carson and Rodrick (2024) note that without layered defences, a single point of entry can result in full compromise. The proposed architecture addresses this by placing controls at each layer: Conditional Access with MFA for identity, Defender for endpoints, Cato SASE with IPS for the network, and Azure Backup with offsite replication for data.

5.2 AI-Driven Security Controls

The table below outlines the proposed security tooling and the capability improvement over current controls.

Current Control	Proposed Control	Capability Improvement
Kaspersky (signature-based)	Microsoft Defender for Endpoint	AI-driven behavioural analysis, automated investigation and response
No visibility across endpoints	Defender XDR	Centralised threat detection across identity, email, endpoints, and cloud apps
No vulnerability management	Defender Vulnerability Management	Continuous vulnerability assessment with risk-based prioritisation

Defender for Endpoint works differently from Kaspersky. Instead of matching files against a database of known signatures, it uses machine learning to spot suspicious behaviour, which is how it would have caught the lateral movement that Kaspersky missed (Carugati, 2023). Defender XDR goes further by correlating signals from identity, email, and endpoint data to flag attack patterns that no single tool would catch on its own. That said, AI-based detection is not something you can set and forget. Someone still needs to review alerts and investigate, whether that is an in-house resource or a managed security partner. We should also keep in mind that AI detection is non-deterministic. The same network behaviour might get flagged one day and not the next. Fine tuning baselines and acceptance of some false positives should be expected early on.

5.3 Data Sovereignty and Regulatory Compliance

Running operations across three countries brings data residency into play. The UAE's Federal Decree-Law No. 45 of 2021 (UAE Government, 2021) sets out requirements for personal data protection, and Azure's UAE North region in Dubai covers the primary workloads. The problem is that Azure has no data centre in Qatar or Bahrain, so data from those locations ends up processed in UAE North. It needs to be clearly documented in the data processing agreements and checked against local requirements in each country (Microsoft, 2024). We also need to consider if the e-commerce platform handles card payments as PCI DSS compliance may apply. Moving to Azure does shift some of that compliance burden to Microsoft, but the business still owns the application-level controls.

5.4 Risk Assessment

The following table summarises the key risks, assessed by likelihood and impact.

Risk	Likelihood	Impact	Mitigation
Vendor concentration (Microsoft)	High	Medium	Terraform portability; annual review
SASE single vendor (Cato)	Medium	Medium	SLA commitments; exit strategy
Data residency non-compliance	Medium	High	Azure UAE North; legal review for Qatar/Bahrain
Migration downtime	Medium	High	Phased migration; rollback per workload
Cloud cost overrun	Medium	Medium	Tagging; Cost Management alerts; RI review at 6 months
Staff skills gap	High	Medium	Managed services partner (12 months); AZ-104 certification
Residual ransomware risk	Low	High	Layered controls; regular DR testing

6.0 Future Recommendations

6.1 Edge Computing at Retail Locations

If the business grows its D2C channels, there could be value in processing inventory and customer data right at the showroom level using Azure IoT Edge, with results feeding back to the cloud. But with only 10 showrooms today, the overhead of managing edge devices probably is not worth it yet. This is one to revisit when the business scales up or starts adding in-store digital experiences.

6.2 AI-Driven Analytics and Operations

There is an opportunity to use Azure AI for demand forecasting and understanding customer behaviour across B2C and D2C. Carugati (2023) makes the point that cloud-AI convergence is opening up analytics capabilities that used to require serious infrastructure investment. The sensible starting point is Power BI dashboards built on the Dynamics 365 data that will already be there, rather than jumping straight to custom machine learning models.

6.3 Serverless for E-commerce Operations

Once the e-commerce platform settles on Azure App Service, it would make sense to look at moving event-driven tasks like order processing and inventory webhooks to Azure Functions. Toosi et al. (2025) make a strong case that serverless works well for workloads with unpredictable traffic because you only pay when code actually runs.

6.4 Multi-Cloud and Emerging Technology Considerations

The dependence on Azure covered in Section 4.0 should be reviewed once the migration is complete. A multi-cloud assessment at the 12 to 18 month mark makes sense, and data export paths should stay open for all critical systems. Quantum computing is not relevant for a business this size today, but quantum-safe cryptography is worth watching as it could change encryption standards in the future.

7.0 Conclusion

The ransomware attack forced the conversation, but the issues at Al Dar Furnishings were there long before that. This report sets out a migration plan that covers all seven gaps by putting controls in place across identity, endpoints, the network, and data. Most of the workloads move to SaaS, which cuts down on the work the IT team has to do day to day. Terraform is there to make sure the business is not permanently locked into one provider. The recommendations at the end are kept practical and based on what the business actually needs right now, not what sounds impressive on paper.

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